



STM32F3 Technical Training

For reference only

Refer to the latest documents for details



STM32F3xx Block Diagrams

STM32F30x Series

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ARM 32-bit Cortex-M4F CPU

Operating Voltage:

- VDD = 2.0 V to 3.6 V or 1.8V +/-8%
- VBAT = 1.65V to 3.6 V

Safe Reset System (Integrated Power On Reset (POR)/Power Down Reset (PDR) + Programmable voltage detector (PVD))

Embedded Memories:

- FLASH: up to 256 KB
- SRAM: up to 40 KB SRAM + 8KB CCM RAM

CRC calculation unit

2 x DMA: 12 channels .

Power Supply with software configurable internal regulator and low power modes.

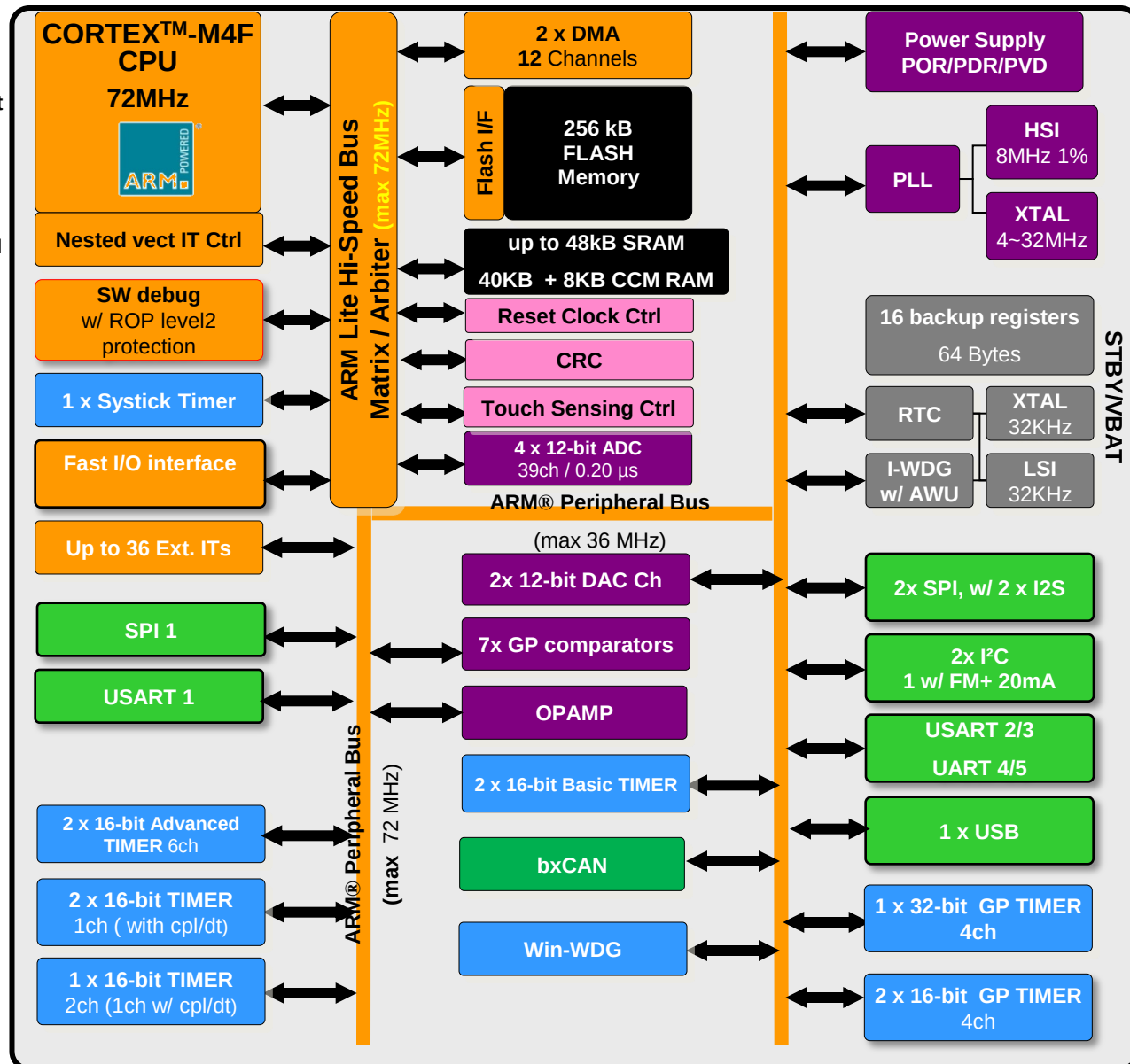
Low Power Modes with Auto Wake-up

Low power calendar RTC with 64 bytes of backup registers

Up to 72 MHz frequency managed & monitored by the Clock Control w/ Clock Security System

Rich set of peripherals & IOs

- 2x 12-bit DAC channels with output buffer
- 7 general purpose comparators (Window mode and wakeup from low-power mode)
- 4 operational amplifiers
- Dual Watchdog Architecture
- 13 Timers w/ advanced control features (including 1 Cortex SysTick timer and 2 WDGs timers)
- 14 communication Interfaces
- Up to 87 fast I/Os all mappable on external interrupts/event
- 4x12-bits 5Mps ADC w/ up to 39 external channels + Temperature sensor/ voltage reference/VBAT measurement



STM32F37x Series

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ARM 32-bit Cortex-M4F CPU

Operating Voltage:

- VDD = 2.0 V to 3.6 V or 1.8V +/-8%
- VBAT = 1.65 V to 3.6 V

Safe Reset System (Integrated Power On Reset (POR)/Power Down Reset (PDR) + Programmable voltage detector (PVD))

Embedded Memories:

- FLASH: up 256 Kbytes
- SRAM: up 32Kbytes

CRC calculation unit

2 x DMA: 12 Channels

Power Supply with software configurable internal regulator and low power modes.

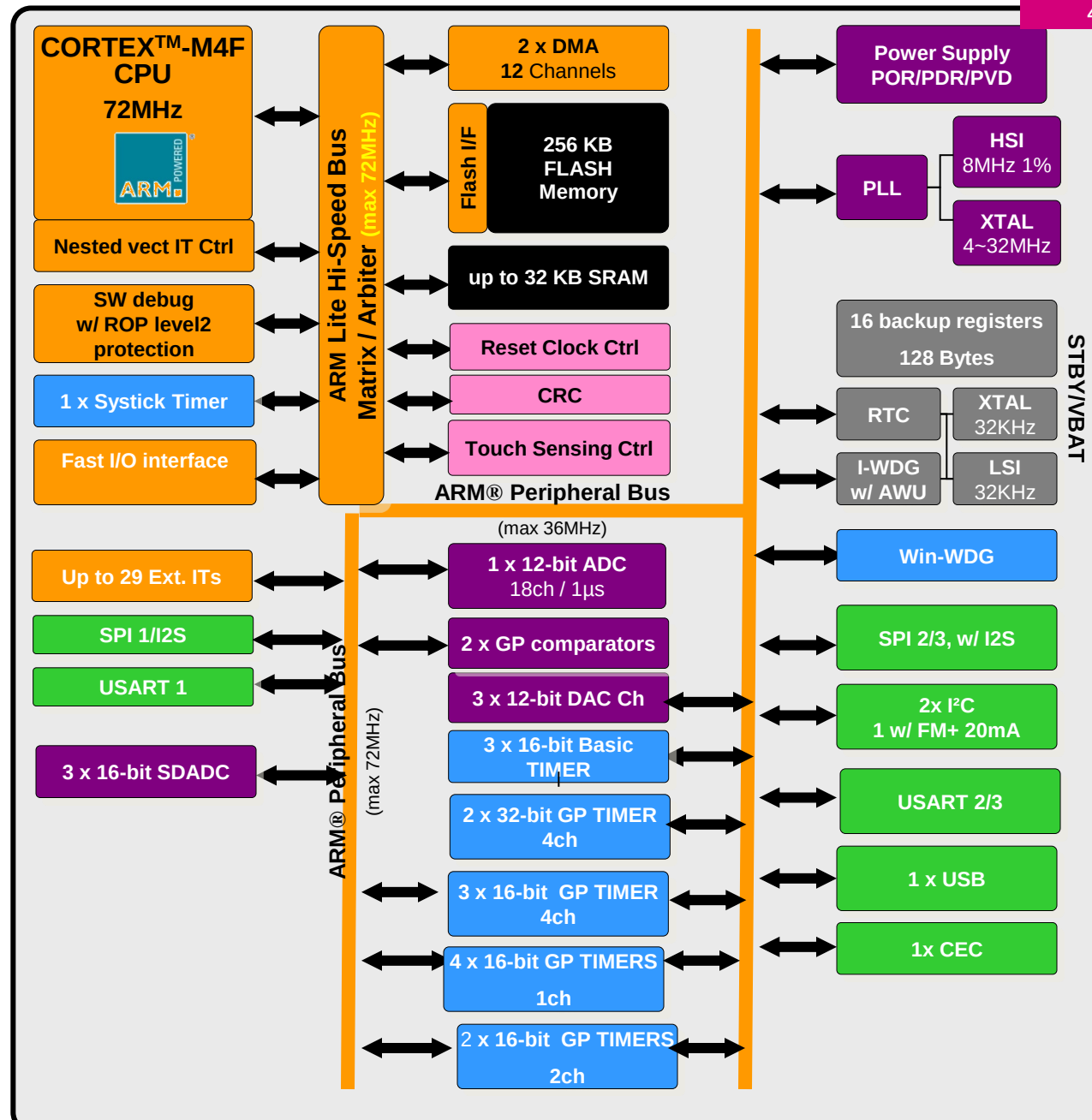
Low Power Modes with Auto Wake-up

Low power calendar RTC with 128 bytes of backup registers

Up to 72 MHz frequency managed & monitored by the Clock Control w/ Clock Security System

Rich set of peripherals & IOs

- 3 x 12-bit DAC channels with output buffer
- 2 general purpose comparators (Window mode and wakeup from low-power mode)
- Dual Watchdog Architecture
- 17 Timers (including Cortex SysTick and WDGs)
- 14 communication Interfaces
- Up to 84 fast I/Os all mappable on external interrupts/event
- 1 x12-bits SAR ADC w/ up to 18 external channels .
- 3 x 16-bit Sigma-Delta ADC with conversion speed up to 50 kcps and up to 19 single/ 10 diff channels



Memory and System Architecture

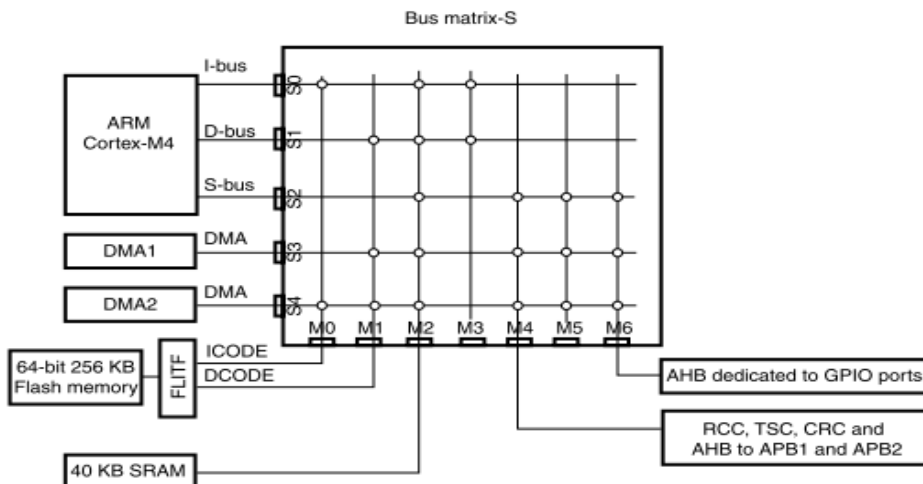
System Architecture

In STM32F30x

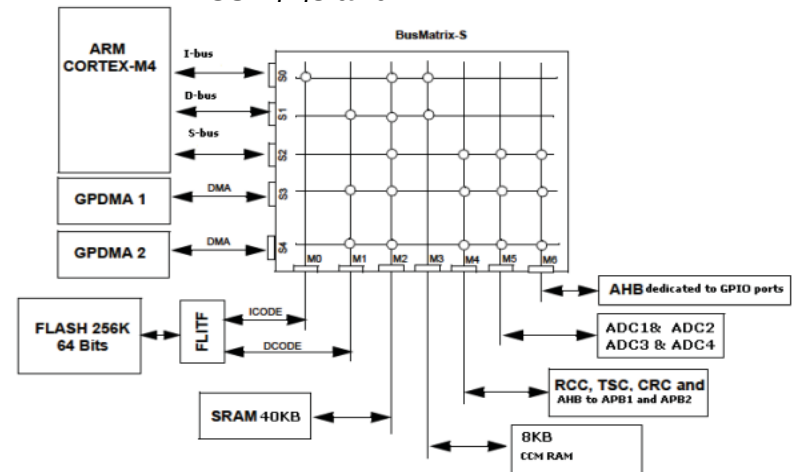
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In STM32F37x

- Five masters:
 - Cortex-M4F core I-bus
 - Cortex-M4F core D-bus
 - Cortex-M4F core S-bus
 - GP-DMA1 and GP-DMA2 (general-purpose DMAs)
- Five slaves:
 - Internal SRAM
 - Internal Flash memory
 - AHB to APBx (APB1 or APB2), which connect all the APB peripherals
 - AHB dedicated to GPIO ports



- Five masters:
 - Cortex-M4 core I-bus
 - Cortex-M4 core D-bus
 - Cortex-M4 core S-bus
 - GP-DMA1 and GP-DMA2 (general-purpose DMAs)
- Seven slaves:
 - Internal SRAM on the Dcode
 - Internal SRAM on the ICode (CCM RAM)
 - Internal Flash memory
 - AHB to APBx (APB1 or APB2), which connect all the APB peripherals
 - AHB dedicated to GPIO ports
 - ADCs 1,2,3 and 4.



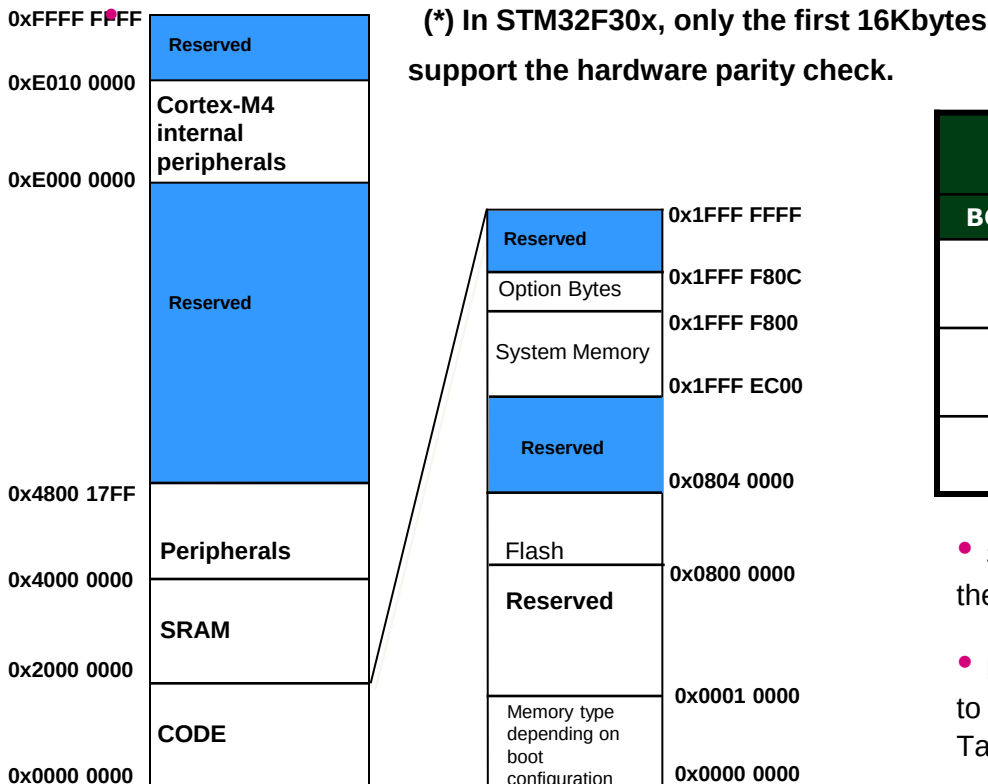
Memory Mapping and Boot Modes

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- Addressable memory space of 4 Gbytes
- FLASH : up to 256 Kbytes
- RAM:
 - Up to 40 (F30x) and 32 (F37x) Kbytes SRAM with HW parity check (*)
 - Up to 8 Kbytes CCM RAM with HW parity check (STM32F30xonly)
 - 4 bits per word for parity check

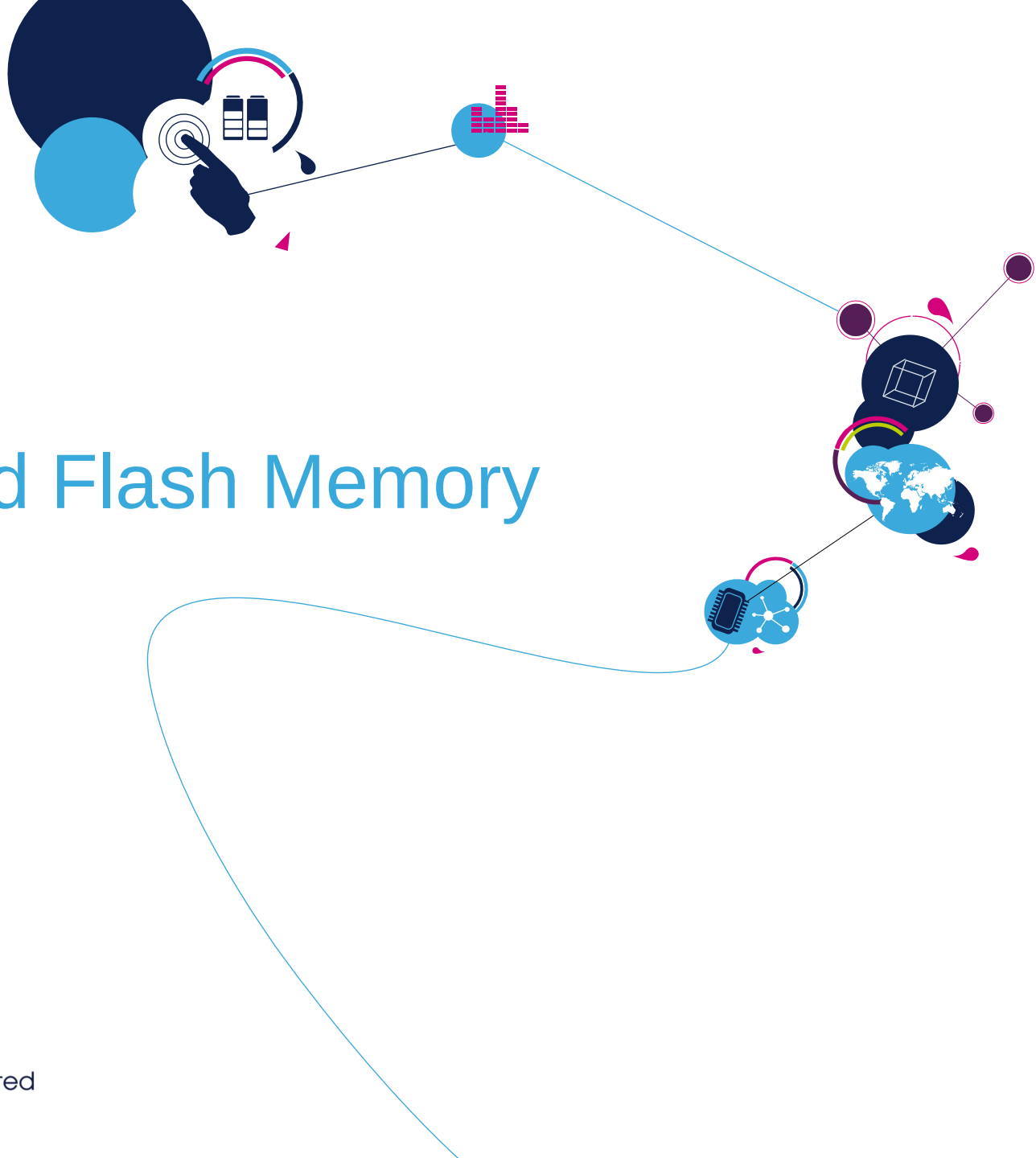
Boot modes

- Depending on the Boot configuration, Embedded Flash memory, System memory or Embedded SRAM memory is aliased at @0x00 thanks to memory remapping bits in SYSCFG registers.
- Even when aliased, these memories are still accessible from their original memory space.
- The boot configuration is defined with **BOOT0 pin** and **BOOT1 bit** in USER Option Byte.



BOOT Mode Selection		Boot Mode	Aliasing
BOOT1	BOOT0		
x	0	User Flash	User Flash is selected as boot space
1	1	System memory	SystemMemory is selected as boot space
0	1	Embedded SRAM	Embedded SRAM is selected as boot space

- **System memory** : contains the Bootloader used to re-program the FLASH through USART or USB .
- **Boot from SRAM** : In the application initialization code you have to Relocate the Vector Table in SRAM using the NVIC Exception Table and Offset register.



Embedded Flash Memory

Flash Features Overview

- Flash general features:
 - Up to 256 KBytes
 - 128 pages of 2KBytes size
 - Access time: 35ns
 - Half word (16-bit) program time: 52.5µs (Typ)
 - Page erase time and Mass erase time: 20ms (Min), 40ms (Max)
- Flash interface features:
 - Read Interface with pre-fetch buffer
 - Option Bytes loader
 - Flash program/erase operations
 - Types of Protection:
 - Readout protection: Level 0, Level 1 and Level 2
 - Write Protection

Flash Memory Organization

- Main memory block containing 128 pages of 2Kbyte each.
- Information block contains the system memory and option bytes, is divided into two parts:
 - **System Memory**
 - 8 KB size
 - contains the bootloader which is used to reprogram the Flash memory through USART1, USART2 or USB.
 - used to boot the device in System memory boot mode.
 - programmed by ST when the device is manufactured, and protected against unwanted write/erase operations.
 - **8 Option bytes** : Can be read from the memory location starting from 0x1FFFF800 or from the Option byte register (FLASH_OBR) in the Flash memory interface register area.
 - 4 for write protection
 - 1 for read protection
 - 1 for device configuration:
 - IWDG HW/SW mode
 - Reset when entering STANDBY mode
 - Reset when entering STOP mode
 - VDDA supervisor
 - BOOT1
 - SRAM parity check
 - 2 For User Data (To store Security IDs, etc.)

Flash Operations (1/2)

- The embedded Flash memory can be programmed using
 - in-circuit programming (ICP) or in-application programming (IAP).
- The Flash program and erase operations are handled by the Flash program and erase controller (FPEC).
 - After reset, the FPEC is **protected against unwanted write or erase operations**. The FLASH_CR register is not accessible in write mode. An unlocking sequence of KEYs should be written to the FLASH_KEYR register to open the access to the FLASH_CR register.
 - The Main Flash can be programmed by writing a half-word (16-bits) at a time. Any attempt to write data that are not half-word long will result in a bus error generating a Hard Fault interrupt.
 - The Main Flash can be erased page-wise or completely (Mass Erase)
 - I-bus stalled during program\erase : Flash cannot be accessed during these operation and have to wait until the BSY bit is reset in the FLASH_SR register to perform the next operation.

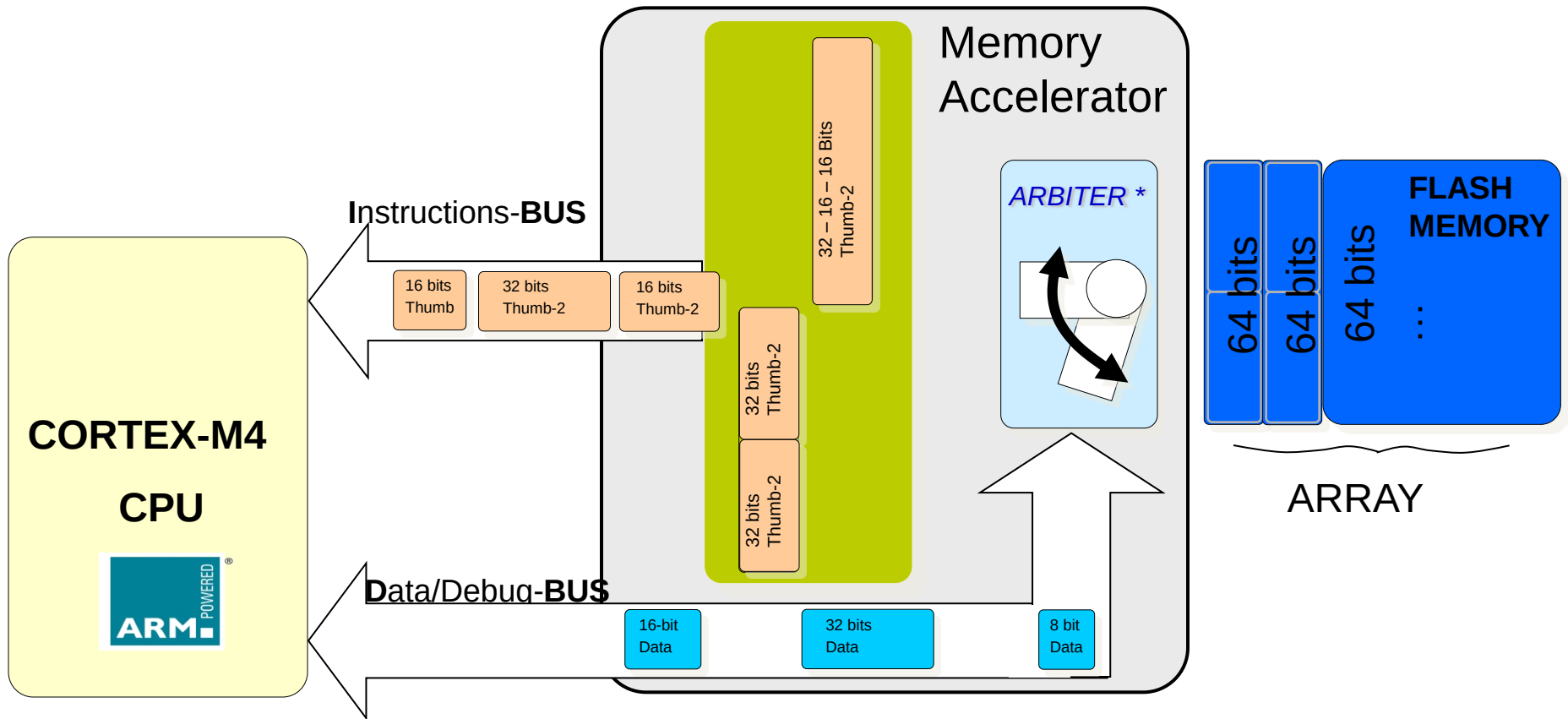
Flash Operations (2/2)

- The **8 option bytes** are programmed differently from main Flash:
 - After unlocking the Flash access, authorize the programming of option bytes by writing same set of KEYS to FLASH_OPTKEYR register to set the OPTWRE bit in the FLASH_CR register.
 - Then set the OPTPG bit in the FLASH_CR register.
 - When the Flash memory read protection option is changed from protected to unprotected, a Mass Erase of the main Flash memory is performed.
 - On POR reset, the option bytes loader performs a read of option bytes and stores the data into the FLASH registers (when programmed, the option bytes are taken into account only after POR reset). User can also use the FORCE_OPTLOAD bit from FLASH_CR register to initiate the option bytes loader (generating SYSTEM reset).
- The Read access can be performed with the following configuration options:
 - Latency: number of wait states for a correct read operation (from 0 to 2).
 - Prefetch buffer of 2x64bit: For faster CPU execution, can be enabled and disabled on the fly.
 - Half Cycle: Flash access can be made on a half cycle of the HCLK to reduce power consumption, enabled by software.

Flash memory prefetch controller

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- **Mission:** Support 72 MHz operation directly from Flash memory
- 64-bits wide Flash with Prefetch (2 x 64bits buffers).



Flash error/status flags and interrupts

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- The Flash program and erase controller provides error and status flags with possible interrupts:
 - **WRPRERR (write protection error flag):** Set by hardware when an address to be erased/programmed belongs to a write-protected part of the Flash memory.
 - **PGERR (programming error flag):** Set by hardware when the data to program is different from 0xFFFF before programming.
 - **EOP (End of operation):** This bit is set by hardware when a Flash or Option byte operation (program/erase) is completed.
 - **BSY (Write/erase operations in progress)**

Interrupt event	Event flag	Enable control bit
End of programming	EOP	EOPIE
Error	WRPRERR PGERR	ERRIE

Flash protections (1/6)

- Two kinds of protections are available:
 - **Write protection** to avoid unwanted writings
 - **Readout protection** to avoid piracy: Level 0, Level 1 and Level 2 (No debug)
 - Both are activated by setting option bytes
- **Write protection**
 - The write protection is implemented with a choice of protecting 2 pages (4K) at a time
 - **4 options bytes** are used to protect all the 256KBytes main Flash program memory
 - Any programming or erase of a protected page is discarded and the Flash will return protection error flag in the FLASH_SR status register
 - **Un-protection**
 - Erase the corresponding bit on WRPx option bytes, x = 0..3.
 - Reset the device (POR Reset) or set the FORCE_OPTLOAD bit to re-load the options bytes for disabling any write protection.
 - The write protection bit values are visible also through FLASH_WRP_R write protection register.

Flash Protections (2/6)

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- **Read protection**

- The read protection is activated by setting the **RDP option byte** and then, by applying POR reset or using FORCE_OPTLOAD bit from FLASH_CR register to reload the new RDP option byte.
- Three levels of protection from no protection (**Level 0**) to maximum protection (**Level 2 or No debug**)

RDP byte value	RDP complement value	Read protection level
0xAA	0x55	Level 0
Any value but 0xAA or 0xCC	Any value (not necessarily complement) but 0x55 and 0x33	Level 1
0xCC	0x33	Level 2 (No debug)

- **Readout protection Level 0**

- No read protection
 - All operations (if no write protection is set) from/to the Flash, option byte or the RTC Backup registers are possible in all boot configurations (Flash user boot, boot RAM, boot loader or debug).

Flash Protections (3/6)

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- **Readout protection Level 1**
 - When this protection is enabled :
 - **User mode:** Code executing in user mode can access main Flash memory and option bytes with all operations.
 - **Debug, boot RAM and boot loader modes:** The main Flash memory and backup registers (RTC_BKPxR in RTC) are totally inaccessible in these modes, a simple read access generates a bus error and a Hard Fault interrupt. Any attempted program/erase operations sets the PGERR flag.
 - **Un-protection:**
 - When the RPD is reprogrammed to the value 0xAA to move back to Level 0, a Mass erase of the main Flash memory is performed and the backup registers (RTC_BKPxR in RTC) are reset.

Flash Protections (4/6)

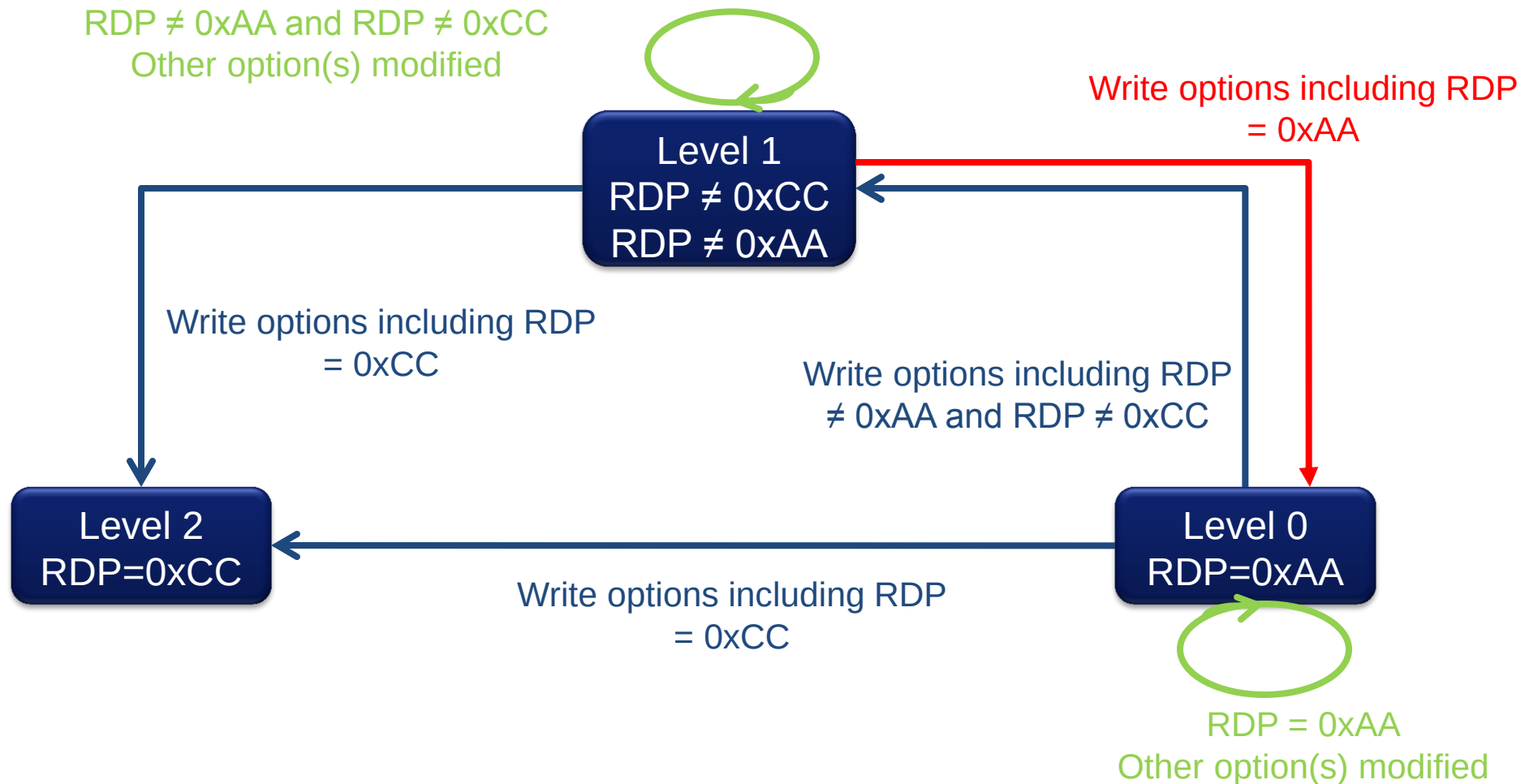
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- **Readout protection Level 2 (No debug)**

- When This protection is enabled :
 - All protections provided by **Level 1** are **active**.
 - **Boot from RAM, boot from system memory and all debug features (serial-wire) are disabled.**
 - Option bytes can no longer be changed except in user mode but not totally ; RDP option byte cannot be programmed/erased and other option bytes can only be programmed (not erased).
- **Un-protection:**
 - Not possible : level 2 cannot be removed at all: it is an **irreversible operation**.

Flash Protections (5/6)

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Option byte write (RDP level increase) includes: Option byte erase and New option byte programming
Option byte write (RDP level decrease) includes: Option byte erase, New option byte programming and Mass Erase
Option byte write (RDP level identical) includes: Option byte erase and New option byte programming

Flash Protections (6/6)

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- Access status versus protection level and execution modes :**

Area	Protection level	User execution			Debug, boot from RAM or boot from system memory (loader)		
		Read	Write	Erase	Read	Write	Erase
Main memory	1	Yes	Yes	Yes	No	No	No
	2	Yes	Yes	Yes	N/A	N/A	N/A
System memory	1	Yes	No	No	Yes	No	No
	2	Yes	No	No	N/A	N/A	N/A
Option bytes	1	Yes	Yes	Yes	Yes	Yes	Yes
	2	Yes	Yes	No	N/A	N/A	N/A
Backup registers	1	Yes	Yes	N/A	No	No	N/A
	2	Yes	Yes	N/A	N/A	N/A	N/A

- List all supported protections and How Enable/Disable them ?

- What is the Maximum Flash Read Frequency?



Thank you !

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